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Numerical Methods for Elliptic Problems – Exercise Sheet 2

Due date: October 27, 2026, 10:15-11:45, Room SP2-0346

Exercise 8:
Let V be a real Hilbert space, $\ell \in V^*$, and $a : V \times V \to \mathbb{R}$ a continuous, positive definite bilinear form, i.e.,

$$|a(u,v)| \leq C\|u\|_V\|v\|_V \quad (u,v \in V), \qquad a(u,u) > 0 \quad (u \neq 0).$$

a) Show that $A : V \to V^*$, defined by $\langle Au, v\rangle_{V^*\times V} := a(u,v)$, is a bounded linear operator and is positive definite in the sense that $\langle Au, u\rangle > 0$ for $u \neq 0$.
b) Show that the variational problem

$$\text{Find } u \in V : \qquad a(u,v) = \ell(v) \quad \text{for all } v \in V$$

is equivalent to the operator equation $Au = \ell$ in V^*.

Exercise 9:
Let V be a real Hilbert space and let $a : V \times V \to \mathbb{R}$ be a symmetric bilinear form satisfying

$$|a(w,v)| \leq C_a\|w\|_V\|v\|_V, \qquad a(v,v) \geq \alpha\|v\|_V^2, \qquad \alpha > 0.$$

Let $V_h \subset V$ be a finite-dimensional subspace, $\ell \in V^*$, and let $u \in V$ and $u_h \in V_h$ solve the continuous and Galerkin problems, respectively. Prove the improved version of Céa's lemma

$$\|u - u_h\|_V \leq \sqrt{\frac{C_a}{\alpha}} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Hint: Use the energy norm $\|v\|_a := a(v,v)^{1/2}$.

Exercise 10:
Let $\Omega \subset \mathbb{R}^2$ be a bounded Lipschitz domain. Assume that, up to a set of boundary measure zero, $\partial\Omega$ is the union of finitely many C^1 pieces Γ_i, $i = 1, \dots, M$. For each piece, assume there is a C^1 diffeomorphism $s_i : Q := (0,1)^2 \to U_i \subset \Omega$ with

$$s_i((0,1) \times \{0\}) = \Gamma_i, \qquad \det(Ds_i) > 0, \qquad \sup_Q \|Ds_i\|_2 + \sup_Q \|(Ds_i)^{-1}\|_2 \leq C.$$

Show that there is a unique bounded linear trace operator $\gamma : H^1(\Omega) \to L^2(\partial\Omega)$ which agrees with the classical boundary restriction for functions in $C^1(\overline{\Omega})$.

Exercise 11:
Let $\Omega = B_1(0) = \{x \in \mathbb{R}^d : |x| < 1\}$, $d \geq 1$, and $u \in H^1(\Omega)$. Define, almost everywhere,

$$\widetilde{u}(x) = \begin{cases} u(x), & |x| < 1, \\ (3 - 2|x|)u(x/|x|^2), & 1 < |x| < 3/2, \\ 0, & 3/2 < |x| < 2. \end{cases}$$

Show that $\widetilde{u} \in H_0^1(B_2(0))$ and prove the estimate

$$\|\widetilde{u}\|_{H^1(B_2(0))} \le C_d \|u\|_{H^1(B_1(0))}.$$

Exercise 12:
Prove the following weighted version of Friedrichs' inequality

$$\int_0^\infty \frac{|u(x)|^2}{(1+x)^\alpha}\,dx \le c_\alpha \int_0^\infty |u'(x)|^2\,dx$$

for $u \in C^1([0,\infty))$ with $u' \in L^2(0,\infty)$ and $u(0) = 0$, and $\alpha > 2$. Does the inequality hold for $\alpha = 2$?

Hint: Use the fundamental theorem of calculus, Cauchy–Schwarz, and Tonelli's (Fubini's) theorem. For the endpoint $\alpha = 2$, try integration by parts on a finite interval $[0, R]$.

Exercise 13:
For a bounded Lipschitz domain $\Omega \subset \mathbb{R}^2$, define the Hilbert space

$$H(\mathrm{curl}, \Omega) := \{\xi \in [L^2(\Omega)]^2 : \mathrm{curl}\,\xi \in L^2(\Omega)\}, \qquad \mathrm{curl}\,\xi := \partial_x \xi_2 - \partial_y \xi_1,$$

equipped with the inner product $(\xi, \zeta)_{H(\mathrm{curl})} := (\xi, \zeta)_{L^2(\Omega)} + (\mathrm{curl}\,\xi, \mathrm{curl}\,\zeta)_{L^2(\Omega)}$, and let $X := H_0^1(\Omega) \times H(\mathrm{curl}, \Omega)$ with its product norm. For $c \ge 0$ and $f \in L^2(\Omega)$, consider the problem of finding $(u, \xi) \in X$ such that, for all $(v, \zeta) \in X$,

$$A((u,\xi),(v,\zeta)) := \int_\Omega (\nabla u - \xi)\cdot(\nabla v - \zeta)\,dx + c\int_\Omega \xi\cdot\zeta\,dx + \int_\Omega \mathrm{curl}\,\xi\ \mathrm{curl}\,\zeta\,dx = \int_\Omega f v\,dx.$$

a) Use the Lax–Milgram lemma to prove existence and uniqueness for $c > 0$. You may use Young's inequality with a suitable parameter.
b) For $c = 0$, show that the bilinear form is not coercive. *Hint:* What is $\mathrm{curl}\,\nabla u$?
c) For $c > 0$, show that the form is not coercive on the stronger space $\widehat{X} := H_0^1(\Omega) \times [H^1(\Omega)]^2$, equipped with its product norm. *Hint:* Use $u = 0$ and $\xi_\varepsilon(x, y) := (\sin(x/\varepsilon), 0)^\top$.

Exercise 14 (Poisson problem on a Pacman domain):
For $0 < \theta < 2\pi$, consider the circular sector

$$\Omega_\theta := \{r(\cos\varphi, \sin\varphi)^\top : 0 < r < 1,\ 0 < \varphi < \theta\}, \qquad \beta := \frac{\pi}{\theta}.$$

We solve $-\Delta u = f$ in Ω_θ with $u = 0$ on $\partial\Omega_\theta$.
a) Let $u(r, \varphi) = (1 - r^2) r^\beta \sin(\beta\varphi)$. Show that for $f = 4(\beta + 1) r^\beta \sin(\beta\varphi)$, u is the solution of the weak form with $f \in L^2(\Omega_\theta)$ and $u \in H_0^1(\Omega_\theta)$.
b) For $\pi < \theta < 2\pi$, show that $u \notin H^2(\Omega_\theta)$. What happens for $\theta = \pi/2$?
c) Starting from `pacman_domain.ipynb`, compute conforming H^1 finite element approximations for `order` $= 1, 2, 3$ and $\theta \in \{\pi/2, 3\pi/2, 7\pi/4\}$ on a sequence of meshes with smaller `maxh`. Plot the L^2 and H^1 errors against the mesh size h(= `maxh`), and estimate the observed rates. For the reentrant sectors, explain why increasing `order` does not recover the smooth-solution convergence rates. Contrast this with $\theta = \pi/2$.

Exercise 15 (Heat conduction with a material inclusion):
Let $\Omega = (0,3) \times (0,2)$ contain the inclusion $D = (1,2) \times (1/2, 3/2)$. Set for $\eta > 0$

$$\kappa = \begin{cases} \eta, & x \in D, \\ 1, & x \in \Omega \setminus \overline{D}, \end{cases} \qquad f = \begin{cases} 1, & x \in D, \\ 0, & x \in \Omega \setminus \overline{D}, \end{cases},$$

and solve $-\operatorname{div}(\kappa \nabla u) = f$ in Ω. Write $\vec{q} := -\kappa \nabla u$ for the physical heat flux and let $\vec{n}$ be the outward unit normal. Impose the boundary conditions with $\alpha > 0$, $u_\infty = 0$, and $g_N = 0$

$$\begin{array}{lll} u = 0 & \text{on } \Gamma_D = \{0\} \times (0,2), & \text{(Dirichlet)}, \\ \vec{q} \cdot \vec{n} = \alpha(u - u_\infty) & \text{on } \Gamma_R = \{3\} \times (0,2), & \text{(Robin)}, \\ \vec{q} \cdot \vec{n} = g_N & \text{on } \Gamma_N = (0,3) \times \{0,2\}, & \text{(Neumann)}. \end{array}$$

a) Derive the weak formulation with nonzero u_∞ and g_N.
b) Implement and simulate the problem starting with `heat_conduction.ipynb`. Use an H^1 discretization of degree $p = 2$ or 3. For $\alpha = 1$ and $\eta \in \{1, 10, 100\}$, plot the temperature and heat flux, compare temperature profiles along $y = 1$, and compute $|D|^{-1} \int_D u \, dx$. Explain the effect of a highly conducting inclusion.
c) For $\eta = 10$ and $\alpha = 1$, compute the signed outward heat flux through each outer edge and compare their sum with

$$\int_{\partial\Omega} \vec{q} \cdot \vec{n} \, ds = \int_\Omega f \, dx = 1.$$

Compare the direct gradient-based flux through the right edge with the Robin boundary law.